

Interpretable Multimodal Deep Ensemble Framework Dissecting Bloodbrain Barrier Permeability with Molecular Features

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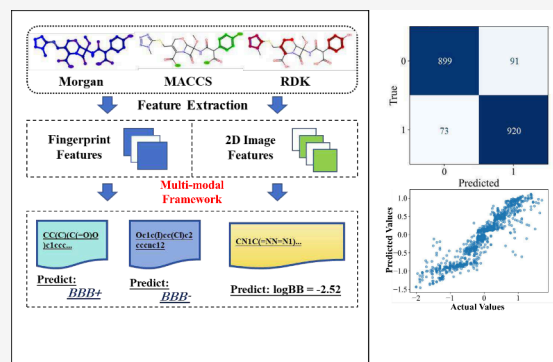


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ABSTRACT: Blood-brain barrier permeability (BBBP) prediction plays a critical role in the drug discovery process, particularly for compounds targeting the central nervous system. While machine learning (ML) has significantly advanced the prediction of BBBP, there remains an urgent need for interpretable ML models that can reveal the physicochemical principles governing BBB permeability. In this study, we propose a multimodal ML framework that integrates molecular fingerprints (Morgan, MACCS, RDK) and image features to improve BBBP prediction. The classification task (BBB-permeable vs nonpermeable) is addressed with a stacking ensemble model combining multiple base classifiers. The proposed framework demonstrates competitive predictive stability, generalization ability, and feature interpretability compared with recent approaches, under comparable evaluation settings. Beyond predictive performance, our framework incorporates Principal Component Analysis (PCA) and Shapley Additive Explanations (SHAP) analysis to highlight key fingerprint features contributing to predictions. The regression task (logBB value prediction) is tackled by a multi-input deep learning framework, incorporating a Transformer encoder for fingerprint processing, a convolutional neural network (CNN) for image feature extraction, and a Multi-Head Attention fusion mechanism to enhance feature interactions. Attention maps derived from the multimodal features reveal token-level relationships within molecular representations. This work provides an interpretable framework for modeling BBBP with enhanced transparency and mechanistic insight and lays the foundation for future studies incorporating transparent descriptors and physics-informed features.



The blood-brain barrier (BBB) is a highly selective semipermeable membrane composed of endothelial cells, which is primarily responsible for regulating the transport of molecules from blood vessels to the central nervous system (CNS).¹ Its strict permeability selectivity effectively protects the brain from harmful substances in the peripheral blood, maintains the homeostatic balance of the brain microenvironment, and ensures the proper physiological function of the brain.² However, the high selectivity of the BBB has emerged as a significant challenge in CNS drug development. Although many drugs have proven to be effective in other tissues, their therapeutic efficacy in treating neurological diseases is often severely limited due to their inability to effectively penetrate the BBB.³ Therefore, an accurate assessment of a drug's BBB permeability during the initial stages of CNS-acting drug design and development is critical, not only for determining whether a compound achieves its therapeutic targets but also for minimizing the side effects of drugs with peripheral activity.^{4,5} The ability of a compound to penetrate the BBB is commonly defined by LogBB, which is the logarithmic ratio of the drug concentration in the brain to the concentration in the blood at equilibrium state.^{6,7} Although the traditional methods for measuring LogBB provide accurate experimental data, they are time-consuming, costly, and complicated in operation,

requiring substantial amounts of purified compounds and extensive animal testing. These limitations make them unsuitable for a high-throughput manner.⁸ As a result, the development of rapid and reliable computational approaches to predict the BBB permeability of candidate drugs has emerged as a critical research direction.^{9,10} In addition, many analysis results in this field have been compiled into data sets and subsequently used to construct BBBP prediction models. For example, the MoleculeNet BBBP data set¹¹ includes 2050 molecules, of which 1567 are BBB permeable (BBB+) and 483 are nonpermeable (BBB-), as well as the recent B3DB data set¹² containing 7807 molecules, of which 4856 are BBB+ and 2851 are BBB-.

With the rapid advancement of computational technologies, machine learning (ML) and deep learning (DL) algorithms have been widely applied in the field of BBB permeability

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prediction. Previous prediction models were primarily qualitative, which predicted whether a query compound was BBB-permeable or impermeable.¹³ Miao et al.¹⁴ employed the DL algorithm in a study on the classification of BBB drugs, which achieved an accuracy of 0.97, an area under the curve (AUC) receiver operating characteristic (ROC) of 0.98, and an F1 score of 0.92. While the results are impressive, the study suffers from a lack of interpretability. Yu et al.¹⁵ developed a protocol to combine the power of ML and DL to generate a set of simple rules that are easy to interpret with high prediction power. This classification model achieved an accuracy of 0.96 and an F1 score of 0.95. Notably, the predictive accuracy of machine learning algorithms is highly dependent on the size of the training data set. To address the issue of small data sets in earlier studies, Shaker et al.¹⁶ created a data set of 7,162 compounds (5,453 BBB permeable compounds and 1,709 BBB nonpermeable compounds), which trained a machine-learning model based on the Light Gradient Boosting Machine (LightGBM) algorithm and achieved an overall accuracy of 89%, an AUC of 0.93, specificity of 0.77 and sensitivity of 0.93. Furthermore, Mazumdar et al.¹⁷ applied balancing techniques such as SMOTE+TOMEK, ADASYN, and SMOTE+ENN to address the class-imbalance issue in the data set. Subsequently, they achieved a prediction accuracy of 97.8% and a ROC-AUC score of 0.98 using a deep neural network model. Although these models have achieved predictions for BBBP, they generally lack transparency, are difficult to explain through local optimization, and have not enabled broader comparative discussions. Recently, Jia and Sosso¹⁸ addressed this interpretability bottleneck by developing a transparent machine learning model for BBB permeability prediction based on a single class of interpretable descriptors-molecular cliques. Trained on the B3DB data set, their random forest model achieved predictive performance comparable to more complex methods (Models such as Deep-B³¹⁹ and LightBBB¹⁶), while providing direct insights into structure-function relationships. Notably, by employing a naive Bayes classifier alongside, they quantified the contribution of individual substructures to permeability, revealing several physicochemical patterns consistent with prior domain knowledge. This study highlights the growing emphasis on explainable models in BBB research and underscores the importance of incorporating interpretable feature representations for mechanistic insight and drug design guidance.

In the last several years, researchers have been attempting to develop quantitative models for predicting LogBB values. Ciura et al.²⁰ developed multilinear regression (MLR) and support vector machine (SVM) models with known LogBB values for 45 marketed drugs. The R^2 values for the SVM and MLR models in the training set were 0.69 and 0.76, respectively. However, given the small size of the data set, the applicability and generalizability of these models may be limited. Wu et al.⁴ utilized a database composed of 300 compounds to develop a quantitative structure-activity relationship (QSAR) model for predicting BBB permeability by combining artificial neural networks (ANN) with molecular structure and property descriptors. The correlation coefficient R of the constructed prediction model, the relative error (RE) and the root-mean-square error (RMSE) were 0.956, 0.857, and 0.171, respectively. To further enhance the robustness and accuracy of predictive models, Shaker et al.¹³ compiled a large data set of 1276 LogBB experimental measurements from literature and proposed the LogBB_Pred quantitative model,

which exhibited superior predictive performance with an R^2 value of 0.61 on an independent test data set. While existing LogBB prediction models provide valuable insights, their limited data set sizes, feature representations, and validation approaches restrict their broader applicability. To overcome these challenges, a proposed model should incorporate larger, more diverse data sets, multimodal feature integration, ensemble learning techniques, and rigorous external validation to ensure higher predictive accuracy, robustness, and real-world applicability.

In this study, we propose a multimodal feature fusion framework for blood-brain barrier permeability (BBBP) prediction, which integrates molecular fingerprints (Morgan, MACCS, RDK) and 2D molecular image-derived features to enrich molecular representations. Molecular data from the B3DB data set are preprocessed through descriptor computation and image generation, followed by dimensionality reduction using principal component analysis (PCA) and interaction feature construction. For the classification task (BBB-permeable vs nonpermeable), we employ a stacking ensemble model composed of multiple base learners to enhance predictive robustness and generalization. For the regression task (logBB value prediction), we design a multi-input deep learning architecture that combines a Transformer encoder for fingerprint sequence encoding, a convolutional neural network (CNN) for image feature extraction, and a multihead attention mechanism for cross-modal feature interaction. Additionally, CatBoost and XGBoost regressors are integrated into the framework through an ensemble learning strategy to further refine regression outputs. The proposed models are evaluated on the B3DB data set using stratified cross-validation and multiple performance metrics under benchmarking protocols aligned with recent work. In addition to predictive performance, we incorporate SHAP value analysis and attention weight visualization to improve interpretability and identify key fingerprint features and structural tokens associated with BBB permeability. Rather than aiming solely for accuracy, this work takes a step toward more interpretable BBBP modeling by combining complementary modalities and explanation techniques. We hope this framework can serve as a useful reference for future efforts incorporating transparent descriptors and physics-informed representations of CNS drug development.

Data Preparation and Pretreatment. Previous works on blood-brain barrier permeability (BBBP) prediction have developed and validated various data sets.^{16,21} In this study, a training data set was constructed based on the B3DB¹² data set, which includes 7807 drug-like small molecules, comprising 4956 “BBBP positive” (BBB+) and 2851 “BBBP negative” (BBB-) entries, and 1058 compounds have LogBB values available. LogBB values represent the ratio of the compound's concentration in the brain relative to the concentration in the blood, serving as an additional measure of BBB permeability, which defined²² as $\log BB = \log (C_{\text{brain}}/C_{\text{blood}})$. Besides LogBB, this data set contains information such as compound name, SMILES string,²³ PubChem compound identifier (CID),²⁴ BBBP type (BBB+/BBB-), International Chemical Identifier (InChI),²⁵ threshold value, and the literature source for the data. As shown in Figure 1(a), we use two key-value pairs such as SMILES-Type (BBBP category) and SMILES-logBB as the training data for the machine learning models.

The performance of BBBP prediction models is closely tied to the molecular representations used to encode molecular

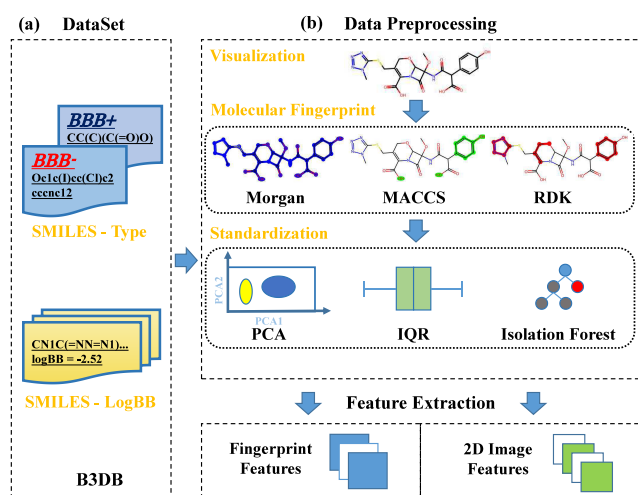


Figure 1. Data set and data preprocessing. (a) B3DB database used as training data set, where classification data is composed of SMILES-Type data pairs and regression data consists of SMILES-LogBB data pairs. (b) Data preprocessing contains visualization, molecular fingerprint, standardization, and feature extraction.

features for similarity assessment in medicinal chemistry.²⁶ In this study, data preprocessing involved several key steps, as shown in Figure 1(b): First of all, 1024 RDKit,²⁷ 1024 Morgan,²⁸ and 166 MACCS²⁹ fingerprint descriptors were computed from SMILES using the RDKit package³⁰ of Python for each compound in the data sets. When considering the prediction of logBB values, it is difficult to achieve the desired results through a single data input, so we adopted multiple sources of data input to expand the scope of feature extraction.¹⁹ A function was implemented transforming

SMILES strings to molecular structures and extracting image features for incorporation into the models. Image features associated with each compound were then resized to 128 × 128 pixels, converting to tensors, and flattening into one-dimensional feature vectors. Principal Component Analysis (PCA)³¹ was then applied to reduce the dimensionality of the normalized fingerprints and image features, retaining the top 30 principal components for both feature sets. Interaction features between the reduced fingerprints and image features were generated using polynomial feature expansion with a degree of 2, focusing on the interaction terms only. Outliers in the data set were detected using interquartile range (IQR)³² and Isolation Forest algorithm.³³ The preprocessed data set contains molecular fingerprint features and image features and was saved as a serialized pickle file for subsequent analysis and machine learning modeling.

Ensemble Learning Algorithm for BBBP Classification. In order to enhance the prediction performance of the BBBP classification, we adopted an ensemble learning approach. This method integrates multiple machine learning models to combine their strengths and achieves excellent overall accuracy and robustness. As shown in the Model section in Figure 2 (a), a range of basic models was implemented, including K-Nearest Neighbors (KNN),³⁴ Logistic Regression (LR),³⁵ Support Vector Classifier (SVC),³⁶ Bernoulli Naive Bayes (BernoulliNB),³⁷ Decision Tree (DT),³⁸ Random Forest (RF),³⁹ Gradient Boosting (GB),⁴⁰ Multi-Layer Perceptron (MLP),⁴¹ CatBoost (CB)⁴² and XGBoost (XGB).⁴³ Each model was tuned for optimal performance using hyperparameter optimization with 5-fold cross-validation.⁴⁴ The optimized models were trained on the preprocessed data set and evaluated using multiple metrics, including accuracy, balanced accuracy, precision, recall (also

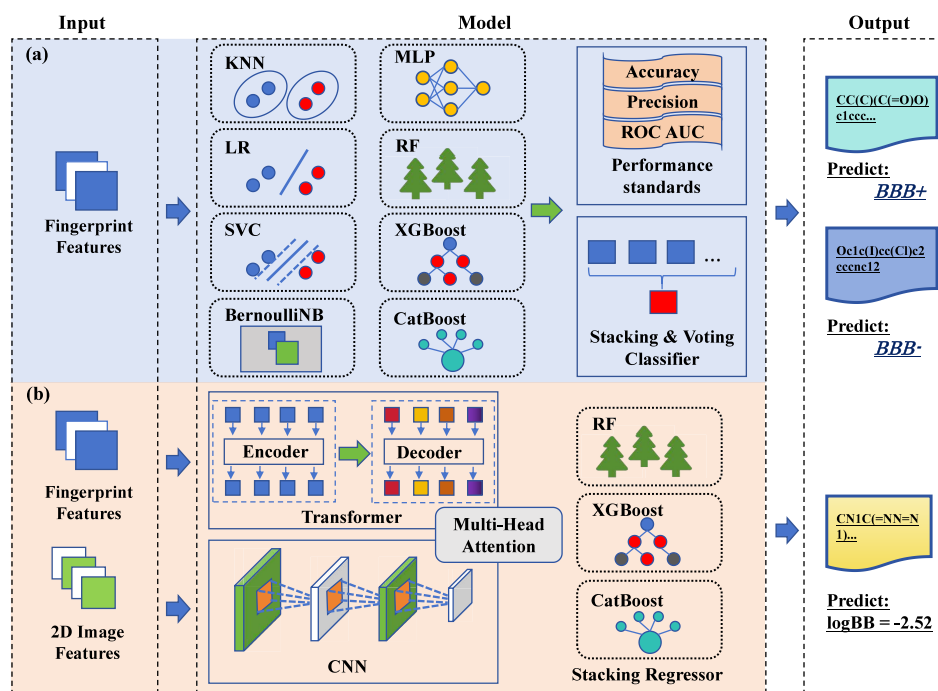


Figure 2. Illustration for the framework of this study. (a) The workflow of ensemble learning algorithm for BBBP prediction containing fingerprint features as input and multiple classification algorithms as the basis for stacking classifier, with a final output of BBB+/BBB- categories. (b) Multi-input algorithm for extracting logBB value, in which fingerprint features are processed by Transformer, image features are processed by CNN, and feature fusion is performed by multihead attention module. The final output is the predicted logBB value obtained by stacking regressor.

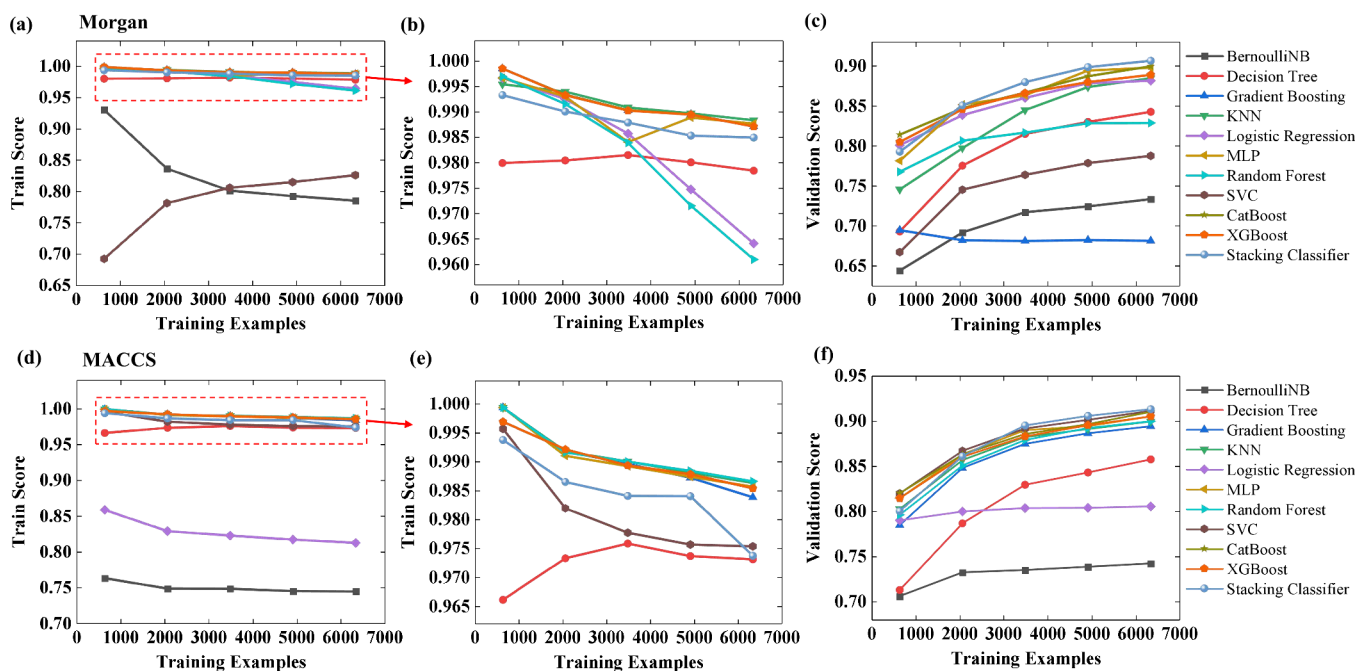


Figure 3. Training performance of models on the B3DB database. (a) The trend of train score with training examples for the models by Morgan molecular fingerprints during the training process, with the overlapping part magnified in (b), while (c) shows the trends in validation score with training examples. (d)–(f) reflect the same model metrics using MACCS fingerprints.

called sensitivity), specificity, F1 score, Matthews correlation coefficient (MCC), Cohen's kappa, and ROC AUC.⁴⁵ Among this The ROC AUC is the area under the curve of the ROC curve, which plots the true positive rate (recall) on the y -axis and the false positive rate on the x -axis. Given TP (True Positive), TN (True Negative), FP (False Positive), FN (False Negative), the formulas for these evaluation metrics as following:⁴⁶

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (1)$$

$$\text{Balanced Accuracy} = \frac{1}{2} \left(\frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} \right) \quad (2)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (3)$$

$$\text{Recall} = \text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (4)$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (5)$$

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (7)$$

$$\kappa = \frac{P_o - P_e}{1 - P_e}, \begin{cases} P_o: \text{observed agreement (i. e., accuracy)} \\ P_e: \text{expected agreement by chance} \end{cases} \quad (8)$$

The ensemble learning method⁴⁷ was constructed using the optimized basic models as individual learners. A stacking classifier is trained based on basic models with weights determined by their ROC AUC scores on the test set. The final estimator was a voting classifier combining RF, GB, CatBoost, and XGBoost using soft voting. The stacking model aggregates the predictions from the basic learners, leveraging their complementary strengths to improve the overall predictive performance. The final output of the classifier is the BBBP type, as shown in Figure 2 (a). To further improve model interpretability, we applied Shapley Additive Explanations (SHAP)^{48–51} to the final ensemble model. SHAP values quantify the contribution of each input feature (e.g., MACCS keys) to the predicted output on a per-sample basis, providing local and global interpretability. This allowed us to identify the most influential molecular fingerprint features driving the BBBP classification results. Visualizing SHAP values not only supports feature-level importance analysis but also offers physicochemically meaningful insights into the structural determinants of BBB permeability.

Multi-input Algorithm for Extracting logBB. To improve the accuracy of logBB prediction, a deep learning framework is developed to integrate molecular fingerprints⁵² and image-based features through a multitask attention fusion mechanism.⁵³ The architecture consists of two modules, as shown in the Model section in Figure 2(b). The first module is a Transformer⁵⁴ Encoder with six layers processes fingerprint features, utilizing a variable number of attention heads based on input dimensionality. A fully connected (FC) layer with ReLU activation maps the Transformer output to a 128-dimensional feature representation. The other module is a Convolutional Neural Network (CNN)⁵⁵ extracting hierarchical representations from molecular images. The network consists of two convolutional layers (32 and 64 channels), followed by max-pooling, flattening, and an FC layer to obtain a 128-dimensional feature vector. The extracted fingerprint and

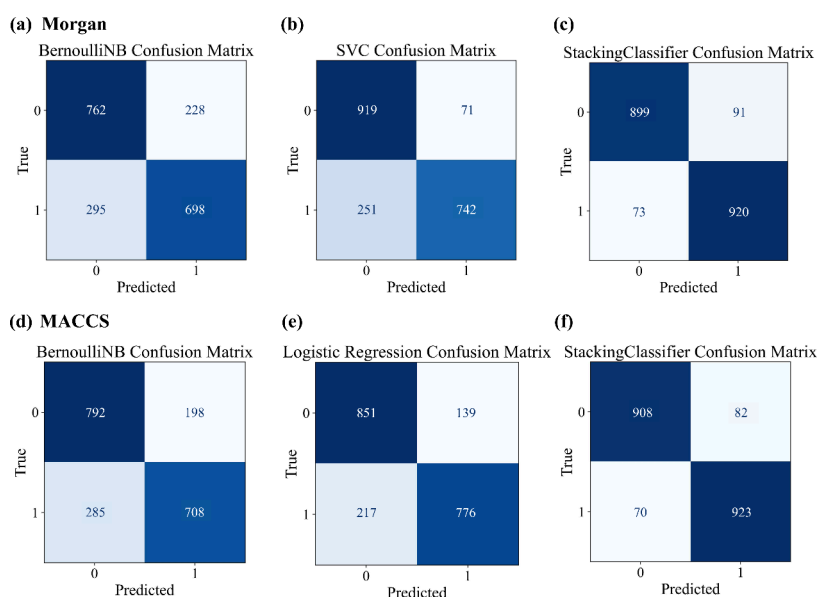


Figure 4. Confusion matrices for models with special points trained on Morgan and MACCS. Panels (a) to (c) represent the confusion matrices of BernoulliNB, SVC and stacking classifier trained on Morgan fingerprints. Panels (d) to (f) display the results of BernoulliNB, LR and stacking classifier using MACCS fingerprints. The labels 0 and 1 in the confusion matrix represent BBB+ and BBB-, respectively.

image features are concatenated and processed through a multihead attention module. Each attention head computes an independent attention score using a fully connected network, and the final feature representation is obtained via softmax-weighted aggregation. A fully connected feed-forward network refines the fused representation through Batch Normalization and ReLU activation, progressively reducing dimensionality from 256 to 1.

To further enhance prediction performance, a stacking regressor is employed, combining the prediction of the model mentioned above and multiple canonical models: RF Regressor, XGB Regressor and CB Regressor. These models served as complementary learners, offering robustness and interpretability in the presence of nonlinear and complex data patterns. A 10-fold cross-validation strategy ensures robust model evaluation, where each fold serves as a test set, while the model is trained on the remaining data. The mean square error (MSE) is used as the optimization objective. The final output of the regressor is the logBB value as shown in Figure 2 (b). In addition to model integration, we incorporated attention weight visualization within the Transformer encoder to further enhance interpretability. The attention mechanism enables the model to assign different weights to fingerprint tokens during prediction, capturing structural dependencies and relative importance among subfeatures. By extracting and visualizing attention maps from trained models, we were able to investigate token-level interactions and highlight which fingerprint dimensions contributed the most prominently to the logBB prediction. This provides complementary insight alongside SHAP-based analyses and supports a more mechanistically grounded understanding of blood-brain barrier permeability.

Due to the consensus that different machine learning models will achieve dissimilar performance on the same batch of data,⁵⁶ this study first discussed the performance of different machine learning models in classification task to categorize input molecule as either BBB+ or BBB-. Meanwhile, feature descriptors play a crucial role in enhancing the effectiveness of

models, which inspires us to utilize different representations to analyze the effectiveness of feature engineering. As shown in Figure 3, we picked Morgan and MACCS molecular fingerprints as the control group, conducting experimental investigation on 10 canonical machine learning models and a stacking classifier based on ensemble learning.

Figure 3 illustrates the training performance of various models trained on Morgan and MACCS molecular fingerprints; we choose two common evaluation metrics such as train score and validation score, testing the variation of these metrics on different training examples. These two metrics of the model are represented as $\text{Train Score/Validation Score} = f(y_{\text{train}}, y_{\text{train}}')/f(y_{\text{val}}, y_{\text{val}}')$, $y_{\text{train}}/y_{\text{val}}$ is the true label of the training/validation set, $y_{\text{train}}'/y_{\text{val}}'$ is the predicted value of the model on the training/validation set.⁵⁷ Figure 3(a) shows the train score of 10 canonical models (KNN, LR, SVC, BernoulliNB, DT, RF, GB, MLP, CB, and XGB) and ensemble learning stacking classifier trained on Morgan fingerprints. The x-axis represents the number of training examples, and the y-axis shows the training score. The plot reveals that most models exhibit a relatively stable or concentrated distribution of training scores across varying sample sizes, with notable exceptions of BernoulliNB and SVC. Figure 3(b) is a zoomed-in version of Figure 3(a), emphasizing the distribution of the training scores. While most models show a compact distribution of scores dropping slightly from 1.0 to 0.96 (KNN and XGBoost perform the best), BernoulliNB stands out with a clear downward trend as the number of training examples increases, indicating that performance deteriorates with larger sample sizes. On the other hand, SVC demonstrates the opposite behavior, with the training score improving as the number of training examples increases, reflecting its adaptability to larger data sets. Figure 3(c) displays the validation score with respect to the number of training examples. Except for GB, which shows a decreasing trend, all of the other models increase in different ranges. Some models with high training scores but significantly lower validation scores are biased toward overfitting, whereas models with low training and

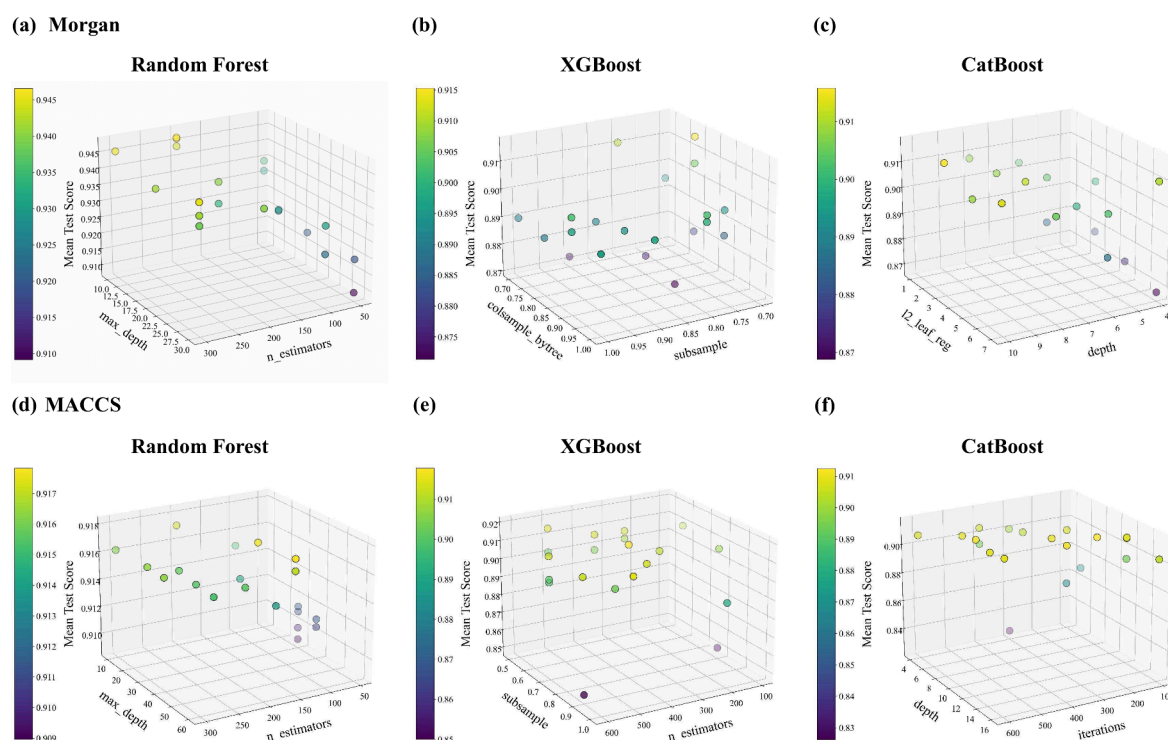


Figure 5. Visualization of hyperparameter tuning. (a)–(c) Hyperparameter search for RF, XGB, and CB across Morgan by RandomizedSearchCV, while the case of MACCS is shown in (d)–(f).

validation scores, such as BernoulliNB, are biased toward underfitting. The stacking classifier, which aggregates the predictive capabilities of multiple models, performs the best in a combined manner, highlighting the advantages of ensemble learning in improving the prediction accuracy during training.

When it comes to Figure 3(d)–(f), these plots reflect the same metrics as Figure 3(a)–(c), but with MACCS fingerprints as the feature input. Similar to the case of Morgan fingerprints, BernoulliNB shows a downward trend in the training scores. A comparable downward trend is also observed for Logistic Regression (LR), although its scores remain generally higher than those of BernoulliNB. Figure 3 (f) presents the validation scores for the models using MACCS fingerprints, which are overall higher than Morgan's case. As in the previous cases, the stacking classifier demonstrated the highest performance, underscoring the robustness and superiority of the ensemble learning approach in both feature scenarios. Finally, Figure 3 also highlights the specific behaviors of individual models, particularly the performance degradation of BernoulliNB with larger training data sets, the performance of DT is modest but stable, and the performance of CB and XGB are among the top performers of all the classical models.

Further analysis of the models' performance is shown in Figure 4. Subplots a–c correspond to the confusion matrices of BernoulliNB, SVC and the stacking classifier using Morgan fingerprints as the feature input, respectively. As observed in Figure 3, BernoulliNB and SVC exhibit different convergence trends in their training scores. This difference is also evident in the confusion matrices. In Figure 4(a), BernoulliNB performs poorly in distinguishing between the two classes, with significant misclassification of both the BBB+ and BBB- labels. The model's inability to correctly classify these instances highlights its performance limitations. In contrast, the SVC

model (Figure 4(b)) shows an improvement in recognizing BBB+ compared to BernoulliNB, which aligns with SVC's better adaptability to larger data sets, as seen in its training performance. However, SVC still struggles with the classification of BBB-, indicating room for improvement. Figure 4(c) reveals the stacking classifier's superior performance, with relatively fewer misclassifications of both BBB+ and BBB- labels. Figure 4 (d) to (f) mirror the confusion matrices of the models trained on MACCS fingerprints, corresponding to BernoulliNB, LR and the stacking classifier. As shown, both BernoulliNB (Figure 4(d)) and LR (Figure 4(e)) exhibit lower classification effectiveness on MACCS fingerprints when compared to the stacking classifier (Figure 4(f)). BernoulliNB continues to perform poorly, with significant misclassifications, while LR shows slightly better results than BernoulliNB but still falls short, corresponding to its training situation. The stacking classifier remains the most effective model, correctly distinguishing between BBB+ and BBB- with minimal misclassification, which validates our choice of ensemble learning and further confirms its robustness across different molecular fingerprint features.

It is worth noting that both Figure 3 and Figure 4 reflect that each model has different performance manifestations based on different molecular fingerprints, with MACCS performing better than Morgan. Therefore, we explored the specific differences in the performance of molecular fingerprints by systematically tuning model hyperparameter. Partial analysis of the parameter tuning scheme is shown in Figure 5. We selected RF (training scores fluctuated and differed significantly between the two features) and XGB and CB (two classical models with the best combined training and validation scores) for our study. These models were optimized within a suitable hyperparameter search space, which includes parameters like "max_depth" (the maximum depth of individual trees, to

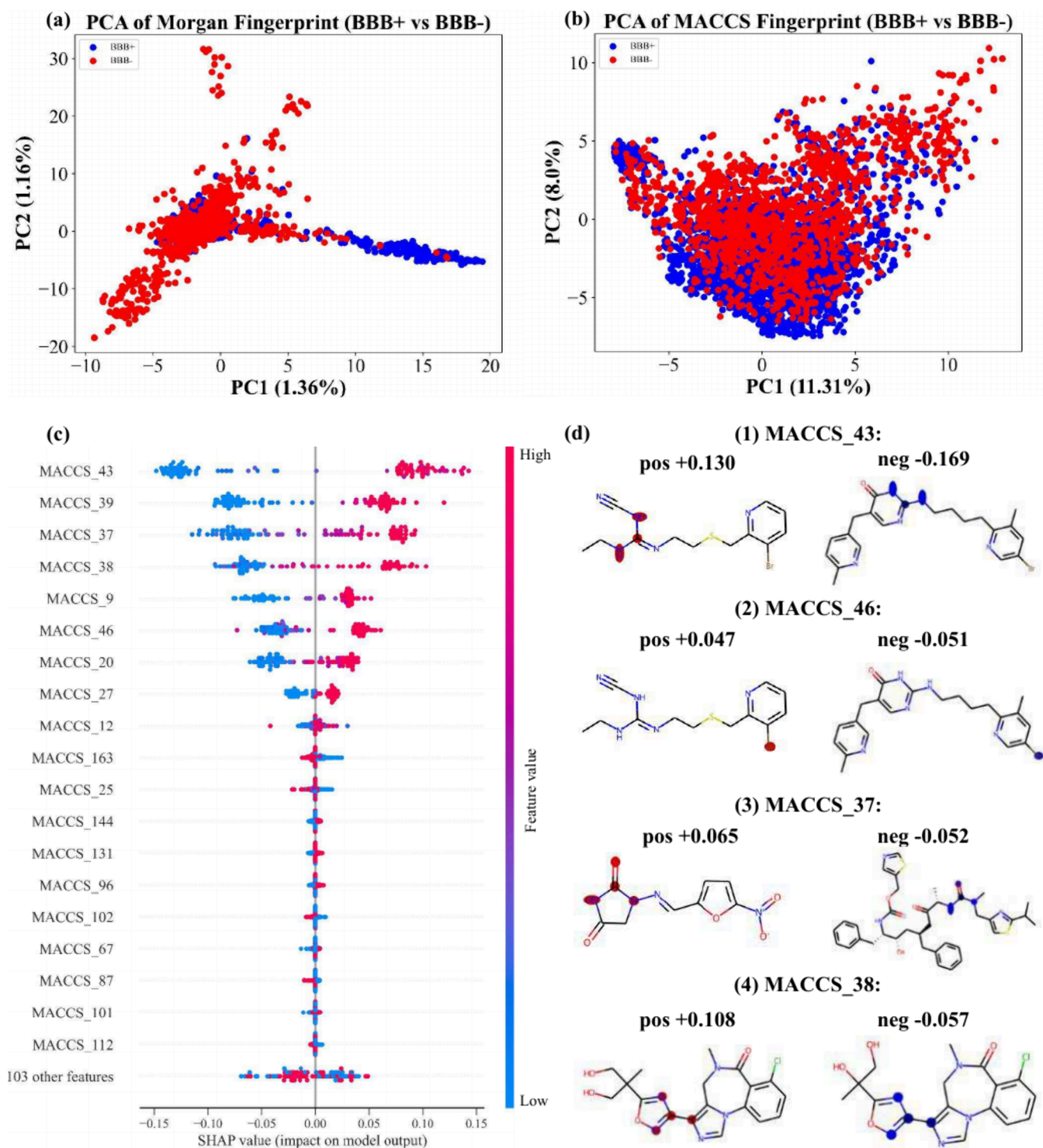


Figure 6. (a) PCA score plot of the first two PCs for BBB+/BBB- with the Morgan features. (b) PCA of the MACCS features. (c) Beeswarm plot illustrating sample-level SHAP value distributions of the model with MACCS. (d) Structural visualization of representative molecules containing substructures annotated with their corresponding SHAP values (pos and neg Represent positive and negative, respectively).

balance model complexity and avoid overfitting), “n_estimators” (the number of trees in the ensemble, to affect model performance and computational cost), “colsample_bytree” (the fraction of features to use when constructing each tree, to reduce overfitting by limiting feature diversity), “subsample” (the fraction of training samples used for each tree, to reduce variance and prevent overfitting), “depth” (the maximum depth of trees, similar to “max_depth”, to control model

complexity), “iterations” (the number of boosting iterations, to influence model accuracy and training time), and “l2_leaf_reg” (regularization parameter that controls the L2 penalty on leaf values, to reduce overfitting by penalizing large coefficients). For RF, as shown in Figure 5(a) and (d), MACCS gives a more balanced distribution of model performance within the parameter space, the mean test score for “max_depth” and “n_estimators” remains relatively stable

across different combinations of these two parameters. This suggests that RF's performance is less sensitive to large variations in the hyperparameter values and when using MACCS, leading to more consistent results.

In the case of XGB, the appropriate parameter space varies according to the fingerprint characteristics. As shown in Figure 5(b,e) for Morgan and MACCS fingerprints, the performance of XGB differs notably. For Morgan, XGB performs best when "colsample_bytree" and "subsample" are set to relatively low values, but other parameter settings are not as good. On the other hand, for MACCS, adjusting "subsample" and "n_estimators" results in a more balanced performance across the parameter space, and the overall performance is still high, as indicated by the more uniform distribution in Figure 5(e). This suggests that XGB performs well with MACCS, but the optimal combination of hyperparameters differs from that for Morgan. When it comes to CB, an interesting situation is observed for the MACCS and Morgan fingerprints. For Morgan, the model performance improves as the "depth" increases and "l2_leaf_reg" decreasing. However, with MACCS, adjusting the "depth" and "iterations" parameters result in better model performance for CB, as shown in corresponding Figure 5(c) and (f). This suggests that CB's performance is more sensitive to the "depth" and "l2_leaf_reg" parameters when using Morgan, while it is more stable and performs better with MACCS when adjusting "depth" and "iterations". The visualized results indicate that different models exhibit varying behaviors depending on the fingerprint feature used. From this phenomenon, we can conclude that further exploration of descriptor engineering and hyperparameter tuning could enhance model generalizability and interpretability.

To explore more intuitively why Morgan and MACCS molecular fingerprints bring different model effects, we further analyze the PCA score plot of the first two principal components (PC1 and PC2) for data sets in Figure 6, evaluating how well the feature sets (Morgan and MACCS) separate the two classes (BBB+ and BBB-) and providing insights into their discriminatory power. PCA reduces the dimensionality of the data by projecting it onto orthogonal axes that capture the maximum variance, and the first two PCs are plotted to visualize the separation between classes in 2D space. As shown in Figure 6(a), there is significant overlap and diffuse clusters, reflecting the Morgan features are not sufficient for distinguishing BBB+ and BBB- compounds. The percentage of variance explained by PCs is indicated in the axis labels, and the low cumulative explained variance suggests that Morgan captures limited data structure. On the contrary, well-separated BBB+/BBB- clusters and higher variance in Figure 6(b) indicate that the MACCS features are effective at differentiating the two classes. The reason why MACCS is superior to Morgan is probably attributed to their characteristics: Morgan represents molecules through cyclic substructure hash encoding, which is informative but has lower interpretability; MACCS adopts 166 predefined substructures, with high computational efficiency and strong interpretability, making it suitable for small sample and similarity analysis tasks in our study. To interpret the contribution of structural features to BBB permeability prediction, SHAP analysis was performed on the model using the filtered MACCS fingerprint as shown in Figure 6(c). The top-ranked features include MACCS_43 (key description: QHAQH. SMARTS⁵⁸ patterns corresponding to the MACCS keys can be found in Json S1 of

Supporting Information), MACCS_39 (key description: OS-(O)O), MACCS_37 (key description: NC(O)N), and MACCS_38 (key description: NC(C)N), exhibited the highest mean SHAP values (e.g., +0.11, +0.07, +0.06, +0.06, respectively), indicating their substantial influence on model output. The whole SHAP values are shown in Figure S5 of Supporting Information. Figure 6(d) shows structural examples containing key substructures with annotated SHAP values, demonstrating positive (pos) and negative (neg) effects of substructures on BBB permeability prediction. As shown in Figure 6(d) (1), among the top contributing MACCS features identified by SHAP analysis, MACCS_43 reflects substructures rich in polar functional groups such as hydrogen bond donors/acceptors and sulfonates (OSO), which are known to reduce BBB permeability⁵⁹ in line with its expected role in increasing polar surface area and limiting passive diffusion across the lipid-rich bloodbrain barrier. As shown in Figure 6(d) (2), MACCS_46 represents brominated substructures, a class of halogens often used to enhance lipophilicity and passive permeability. However, its negative SHAP value here suggests that excessive halogenation may increase molecular weight or introduce metabolic liabilities, thus diminishing overall permeability in this data set context. Further structural analysis was conducted using representative molecules that simultaneously contain MACCS_43 and MACCS_46 substructures; MACCS_43 yields a positive SHAP value (+0.130) and a negative SHAP value (0.169) in two molecules, indicating a much higher contribution than MACCS_46 to predicting a BBB+ label. MACCS_37, associated with secondary amine or amino acid-like fragments, displayed a moderately negative effect (SHAP = -0.052). These groups may participate in hydrogen bonding, potentially reducing permeability, depending on their positioning within the molecular scaffold. Both MACCS_37 and MACCS_38 in Figure 6(d) (3) and (4) are associated with nitrogen-containing motifs such as amines or amino acidlike substructures, exhibiting context-sensitive SHAP values. For instance, MACCS_38 was found to contribute positively in some molecules (e.g., +0.108) but negatively in others (e.g., 0.057), suggesting that symmetric diamine structures may have dual pharmacokinetic implications depending on molecular topology and surrounding chemical environment. These findings are consistent with established pharmacological principles⁶⁰ and emphasize the importance of interpreting substructural contributions in context. While certain motifs may generally enhance or hinder BBB permeability, their net effect is modulated by neighboring groups and overall molecular configuration. This SHAP-guided analysis provides actionable insights into structure–function relationships that could inform CNS-targeted drug design.

Table 1 compares the performance of our model with those of several publicly reported methods on the task of BBB permeability classification. We designed the comparative structure by referring to the evaluation framework used by Jia and Sosso,¹⁸ which reports key metrics such as sensitivity (SN), specificity (SP), MCC, and AUC across models under 5-fold cross validation. The LightBBB and Deep-B³ models, while well-established, report lower specificity (55–64%) and MCC values (0.43 and 0.49, respectively), indicating challenges in correctly identifying the negative class. In contrast, our model achieves both high sensitivity (94%) and specificity (93%), with a resulting MCC of 0.86 and an AUC of 0.98. These results reflect the improved balance between recall and precision, which is critical for real-world applications. We

Table 1. Performance Comparison of Our Model to Publicly Available Models

model	SN (%)	SP (%)	MCC	AUC
LightBBB ¹⁶	86	55	0.43	n/a
Deep-B ³¹⁹	85	64	0.49	0.83
LDA ⁶¹	n/a	n/a	0.60	0.92
XGBoost with mol2vec ⁶²	94	80	n/a	0.96
cliques ¹⁸	90	84	0.74	0.95
cliques (without SMOTE) ¹⁸	94	78	0.74	0.95
Our study	94	93	0.86	0.98

particularly acknowledge the clique method which demonstrated excellent performance (e.g., SN: 90–94%, SP: 78–84%, MCC: 0.74) using interpretable graph-based descriptors. Their work also systematically examined the impact of data sampling strategies, such as SMOTE, and showed that synthetic oversampling had a minimal effect on MCC and AUC. This observation aligns with our own understanding, as SMOTE introduces interpolated data rather than new structural diversity. While our model yields numerically higher performance on several metrics, we attribute this not solely to algorithmic complexity but rather to the combination of rigorous feature engineering, consistent data preprocessing, and model calibration. Importantly, our approach is grounded in interpretable substructure-level fingerprints, which complement the insights provided by more complex models. Overall, our results support the growing recognition that carefully designed feature representations, even with standard classifiers, can match or exceed the performance of more elaborate pipelines in specific cheminformatics tasks.

Next, we turn to the multi-input algorithm for logBB prediction as mentioned above. This algorithm integrates molecular fingerprints and image-based features through a multihead attention fusion mechanism, followed by a stacking ensemble approach for enhanced prediction accuracy. Figure

7(a) to (c) illustrate the training process of the feature fusion part of this model under three molecular fingerprints (Morgan, MACCS and RDK). The training and validation loss for the MACCS model decreases the fastest, with both training and validation losses reaching their lowest values after fewer epochs. This suggests that MACCS, with its more compact and highly informative feature set, is particularly well-suited for this deep learning framework, allowing the model to converge more quickly and efficiently. Then the model trained on Morgan shows a slower decrease in both training and validation loss compared with MACCS. While it eventually reaches a reasonable level of convergence, the loss values are generally higher throughout the training process. This indicates that the model requires more epochs to effectively learn from Morgan fingerprints, which may be due to the larger and more complex feature space that Morgan fingerprints encompass. At last, the RDK fingerprint-based model shows trends similar to those of the Morgan fingerprint model, with a slower reduction in training and validation loss. The final loss is higher than those achieved with MACCS, suggesting that the RDK fingerprint might not be as immediately effective for the task at hand, potentially due to their less specialized and more general feature representation. In addition to the feature fusion part, we adopt integrated learning approach by adding CB and XGB to inject more decisions into the multi-input model, resulting in training process represented in Figure 7(d) and (e). Although the performance of each individual is not excellent, their role in the overall performance of the model is irreplaceable. In our tests, the addition of integrated learning improved the model's performance by 20%.

In Figure 7(f) and Figure 8, we demonstrate PCA score and variance interpretability for individual image features, molecular fingerprint features (Morgan, MACCS, RDK), and their fused features, to observe whether the two features can be well combined in a multi-input model. From Figure 7 (f), we can see that the explained variance of image features is very low

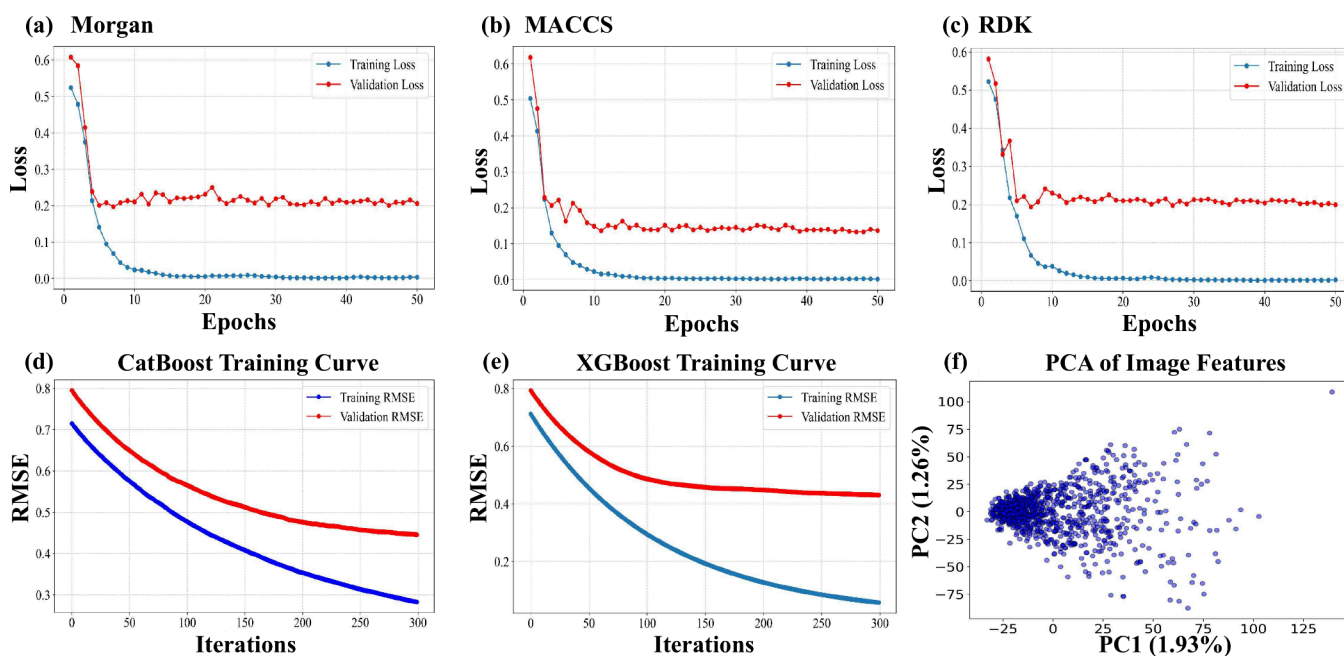


Figure 7. Training process of each module in multi-input algorithm for logBB prediction. (a)–(c) The training loss and validation loss of feature fusion part trained on three molecular fingerprint types: Morgan, MACCS, and RDK. (d) Training and validation of RMSE of CB. (e) The training and validation RMSE of XGB. (f) PCA of image features for logBB.

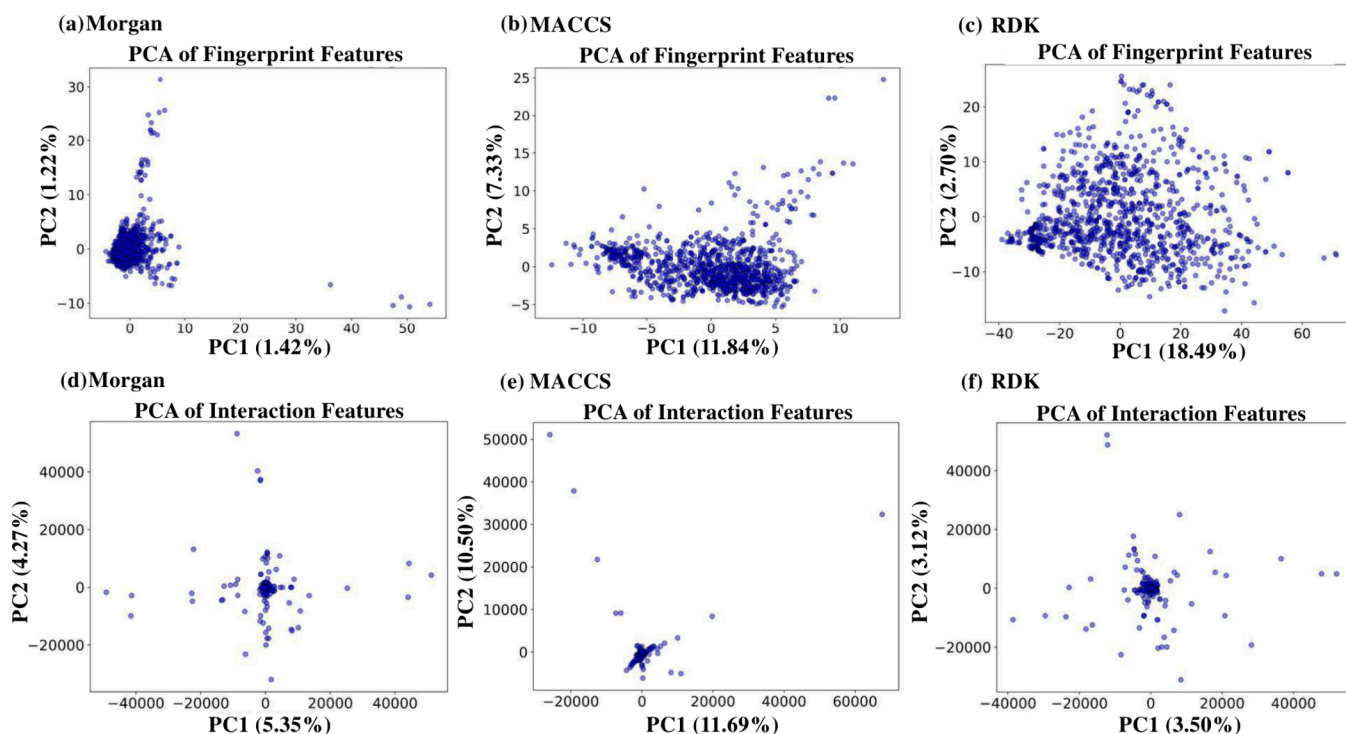


Figure 8. PCA score plot of the first two PCs for logBB. (a)–(c) The PCA of the Morgan, MACCS and RDK features, respectively. (d)–(f) The PCA of the Morgan + image, MACCS + image, and RDK + image, respectively.

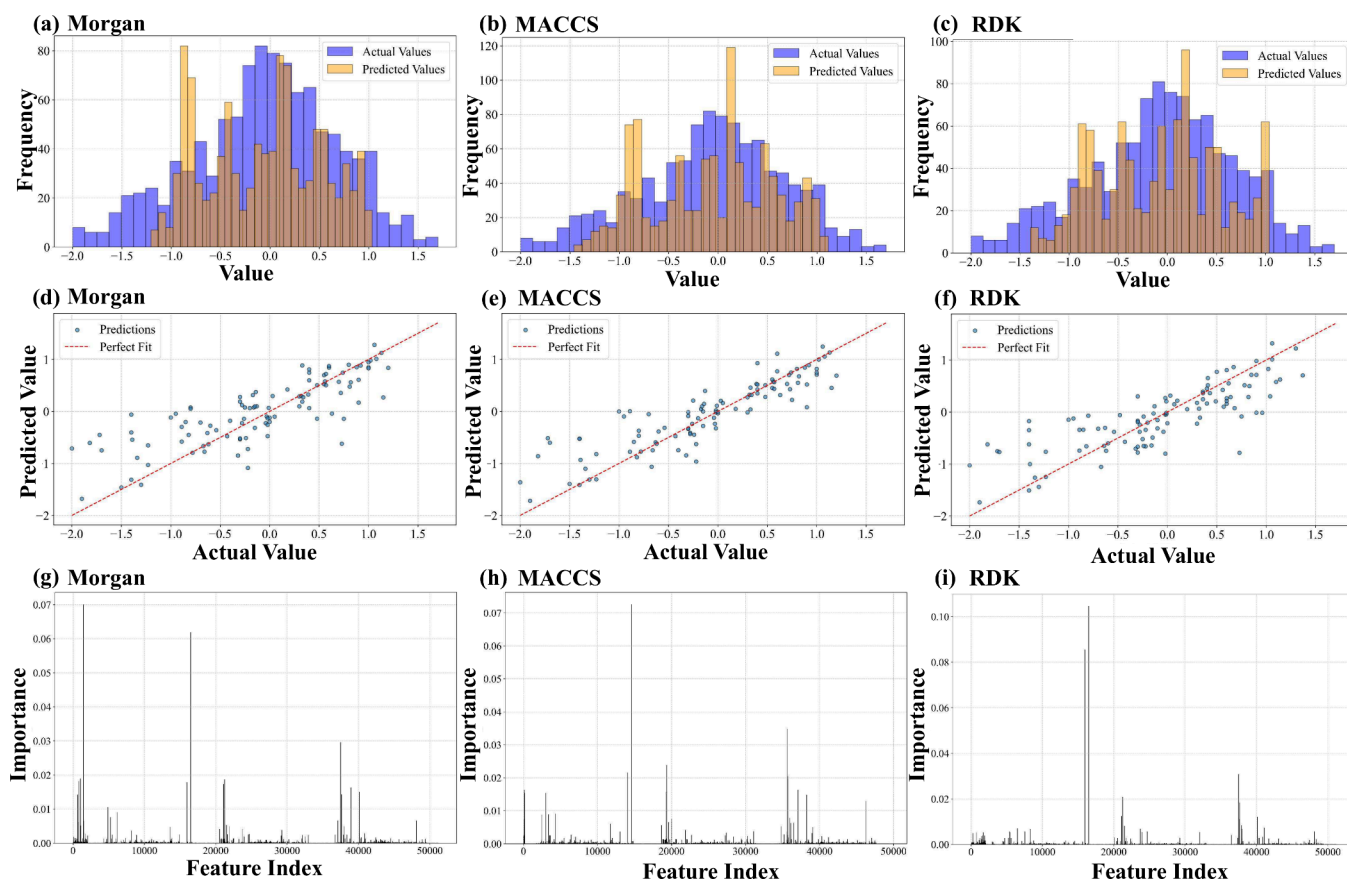


Figure 9. Performance of our multi-input deep learning framework predictions based on Morgan + image, MACCS + image, and RDK + image. (a)–(c) The frequency distribution. (d)–(f) The bivariate scatter of actual and predicted data. (g)–(i) The distribution of feature importance.

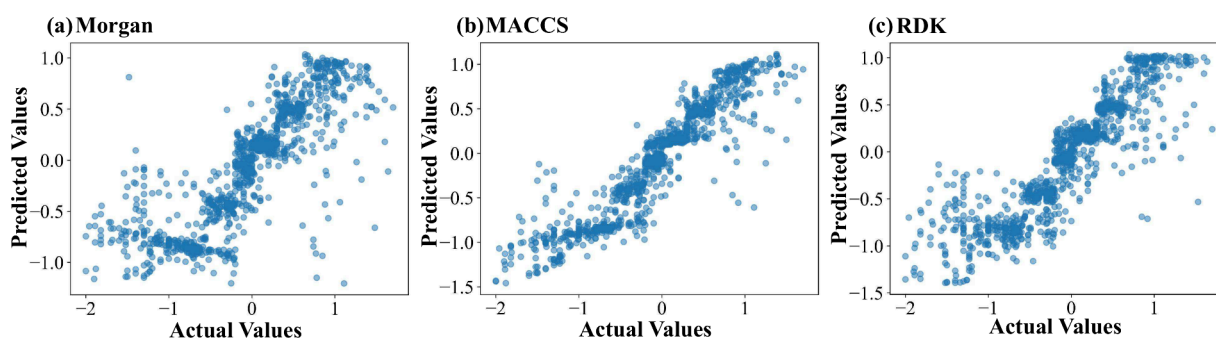


Figure 10. Prediction effects on the B3DB data set. (a) For Morgan: $R^2 = 0.66$, $MSE = 0.1786$. (b) For MACCS: $R^2 = 0.85$, $MSE = 0.0788$. (c) For RDK: $R^2 = 0.78$, $MSE = 0.1189$.

Table 2. Performance Comparison of Our Model to Publicly Available Quantitative Models

Source	compounds	Model	R^2	MSE	Q^2
Wang et al. (2015) ⁶³	439	Mixed (RF, SVM, KNN)	0.64	n/a	n/a
Dyabina et al. (2016) ⁶⁴	529	ANN	0.56	0.41	0.82
Ciura et al. (2020) ²⁰	45	MLR	0.76	n/a	0.73
		SVM	0.69	n/a	0.72
Shaker et al. (2023) ¹³	1276	GBDT	0.59	0.22	n/a
Our study	1058	Multi-input with MACCS	0.85	0.08	n/a

(PC1+PC2 = 3.19%), which means that PCA did not capture effective information structures due to high dimensionality and sparse information. The PCA results of Morgan fingerprints (PC1+PC2 = 2.64%) in Figure 8(a) are worse than the image features, indicating that hash encoding leads to a more uniform distribution of features and PCA cannot effectively reduce dimensionality. MACCS fingerprints' results (PC1+PC2 = 19.17%) in Figure 8(b) are significantly better than those of images and Morgan fingerprints, indicating that their fixed 166 bit structure is easier to parse than hash encoded Morgan fingerprints. As shown in Figure 8(c), the explained variance of RDK fingerprints (PC1+PC2 = 21.19%) is the highest, and the high contribution of PC1 (18.49%) but low contribution of PC2 (2.70%) suggests that there is a dominant direction of data distribution. When combining fingerprint features with image features, the PCA interpretation of Morgan fingerprints increased, but the overall variance remained low (9.62%) as shown in Figure 8(d); MACCS still maintained a high variance interpretation (22.19%) in Figure 8(e), but RDK's variance fused with image features decreased significantly (from 21.19% to 6.6% of Figure 8(f)). As seen above, the optimal feature combination is the MACCS + image, and the Morgan fingerprint has the worst performance, whereas the RDK fingerprint for feature fusion shows information loss, which is in line with the later performance.

Our models are trained on the three fused features mentioned before, and the performance of the models is illustrated through frequency distribution, bivariate scatter, and feature importance plot, highlighting the different trends in predictive behavior. As frequency distribution shown in Figure 9(a) to (c), the model trained on Morgan fingerprints has a narrow distribution of predicted values and a wide distribution of true values, which may result in overfitting, and the model cannot generalize to unseen data. The predicted values of the RDK based model are too scattered compared to the true values, which may result in underfitting and unstable model predictions. Compared with the other two models, the MACCS model has a close distribution of predicted and true

values, indicating that the model can fit the data well. Similar situations occur in the bivariate scatter plots referred to in Figure 9(d)–(f), which visually demonstrate the consistency between predicted and actual values and check the fitting effect. Among them, it can be seen that the data points presented by the models of Morgan and RDK deviate significantly from the diagonal, and the point cloud is relatively loose, which possibly due to nonlinear errors and weak model generalization ability.

When analyzing the feature importance of each fingerprint type by using Random Forest (RF), we observe distinct patterns. As shown in Figure 9(g), the importance of features related to the Morgan fingerprint is relatively scattered, indicating that its hash encoding may lead to information homogenization and difficulty in highlighting key patterns. Some of these features are more important, possibly because the model has learned key local molecular structures from cyclic fingerprints. As shown in Figure 9(h), the feature importance is concentrated in a few key features, indicating that MACCS fingerprints provide more intuitive chemical information and may be more effective than Morgan fingerprints in prediction tasks. In Figure 9(i), except for a very small number of features with extremely high importance, the importance of the remaining features is more balanced but it has not reached the best. This may be because many global molecular attributes may not be related to the target variable, resulting in an increase in feature dimensions but insufficient information.

Figure 10 presents the prediction performance of our multi-input deep learning framework on the B3DB data set. The evaluation metrics were measured based on the coefficient of determination (R^2) and mean squared error (MSE), providing insights into the predictive accuracy and error of each fingerprint type. The R^2 score close to +1 and MSE score close to 0 represent a higher accuracy and better performance. As shown in Figure 10(a), the model trained on Morgan fingerprints achieves an R^2 value of 0.66 and an MSE of 0.1786. This indicates that although the Morgan-based model captures

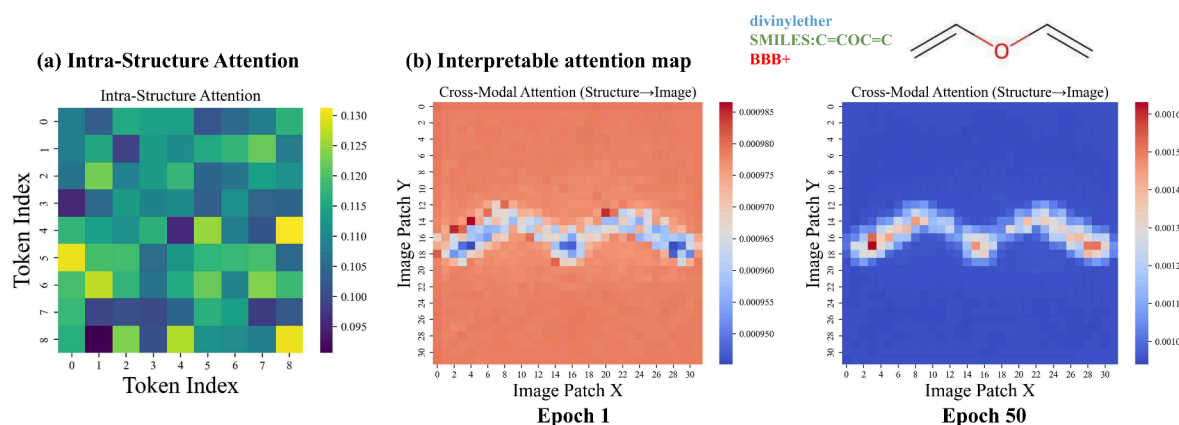


Figure 11. Visualization of attention weights. (a) The attention heatmap representing the internal structure of the fingerprint. (b) The evolution of attention weights from Epoch 1 to Epoch 50 using a cross modal attention heatmap combined with image features.

some of the variance in the data, there is still considerable room for improvement in terms of predictive accuracy. The moderate R^2 suggests that Morgan fingerprints may not fully capture all the nuances required for precise predictions in this context. The model utilizing MACCS fingerprints performs significantly better in Figure 10(b), with a R^2 of 0.85 and an MSE of 0.0788. This demonstrates a strong correlation between predicted and true values, indicating that MACCS fingerprints provide more accurate and reliable predictions. The low MSE further suggests that the MACCS-based model has minimized prediction errors, making it the most effective among the three fingerprints for this data set (which can also be proved in Table 2). The RDKit-based model in Figure 10(c) shows good performance with a R^2 of 0.78 and an MSE of 0.1189. While the R^2 is lower than that of the MACCS model, the RDKit model still demonstrates a solid predictive ability, with an MSE lower than that of the Morgan model. This suggests that RDKit fingerprints offer a balanced prediction, although not as precise as MACCS fingerprints. Our model outperforms previous work at all levels, demonstrating the rationality and robustness of our model architecture.

To further interpret the contribution of different input modalities to logBB prediction, we extracted and visualized cross-modal attention maps from the trained multi-input model. Figure 11(a) illustrates the intrastructure attention maps to understand how the model attends to different spatial or structural aspects during inference. As shown in Figure 11(b), the attention weights between molecular fingerprint tokens and CNN-derived image features reflect how the Transformer encoder gradually sharpens its focus on specific fingerprint dimensions. The evolution of attention weights from Epoch 1 to Epoch 50 highlights the model's increasing reliance on chemically meaningful subfeatures, such as polar or unsaturated fragments, as learned from data. Notably, higher attention scores were observed in regions corresponding to visually distinct functional groups (e.g., double bonds or heteroatoms), suggesting that the image encoder contributes complementary spatial cues that reinforce fingerprint-based reasoning. As a representative example, we examined the molecule divinylether (SMILES: C=COC=C), a BBB-permeable (BBB+) compound. In this case, high intraattention scores were observed around fingerprint dimensions corresponding to alkene and ether functionalities, while the cross-modal attention maps highlighted image regions near the C=C and C-O-C substructures. This alignment suggests that the

model has learned to associate both fingerprint-level and spatial features with physicochemical motifs relevant to the BBB permeability. Together, these attention-based visualizations offer intuitive and interpretable insights into both the internal feature interactions and the cross-modal fusion process, supporting a more transparent understanding of the BBB permeability prediction.

In conclusion, in this study, we present an interpretable multimodal ensemble learning framework for bloodbrain barrier permeability (BBBP) prediction, which integrates molecular fingerprint features (Morgan, MACCS, RDKit) with image-based representations to support both classification and regression tasks. Our approach combines stacking ensemble models with a deep learning architecture that includes a Transformer encoder, convolutional neural network (CNN), and multihead attention fusion, achieving strong performance on the B3DB data set. For BBBP classification, the stacking classifier achieved an AUC of 0.98 and an MCC of 0.86, outperforming several existing baselines. For logBB regression, the multi-input model incorporating MACCS fingerprints and image features attained an R^2 of 0.85 and an MSE of 0.0788, demonstrating strong predictive capability. Furthermore, we incorporated SHAP value analysis and attention weight visualization to enhance interpretability, enabling the identification of key substructures such as halogenated groups, polar motifs, and unsaturated fragments that influence BBB permeability. The visualization of intra- and cross-modal attention weights offered structural-level insight into how the model associates both symbolic and spatial features with the BBB transport properties. Compared with existing deep or boosting models, our method emphasizes interpretability without compromising predictive performance. These results support the growing consensus that the fusion of diverse molecular modalities, when combined with transparent feature attribution techniques, can offer both accurate and mechanistically meaningful predictions. Future work will focus on incorporating more advanced feature selection techniques (e.g., LASSO, SHAP-RFE), expanding external validation across additional data sets, and integrating experimental assays to bridge the gap between computational prediction and pharmacological reality. We hope that the proposed framework can contribute to the design and prioritization of CNS-active compounds in early stage drug discovery.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpclett.5c01077>.

Code data: open source and available on GitHub (<https://github.com/FengDushuo/BBBP-Multi-Modal-Deep-Ensemble-Framework>)

More detailed discussions about algorithm's performance (PDF)

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Notes

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